

(b) The square root of the $\text{Var}(X)$ is called the *standard deviation* of X , and we denote it by $\text{SD}(X)$. That is,

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Discrete random variables are often classified according to their probability mass functions. In the next few sections, we consider some of the more common types.

6 The Bernoulli and Binomial Random Variables

Suppose that a trial, or an experiment, whose outcome can be classified as either a *success* or a *failure* is performed. If we let $X = 1$ when the outcome is a success and $X = 0$ when it is a failure, then the probability mass function of X is given by

$$\begin{aligned} p(0) &= P\{X = 0\} = 1 - p \\ p(1) &= P\{X = 1\} = p \end{aligned} \tag{6.1}$$

where $p, 0 \leq p \leq 1$, is the probability that the trial is a success.

A random variable X is said to be a *Bernoulli random variable* (after the Swiss mathematician James Bernoulli) if its probability mass function is given by Equations (6.1) for some $p \in (0, 1)$.

Suppose now that n independent trials, each of which results in a success with probability p or in a failure with probability $1 - p$, are to be performed. If X represents the number of successes that occur in the n trials, then X is said to be a *binomial random variable* with parameters (n, p) . Thus, a Bernoulli random variable is just a binomial random variable with parameters $(1, p)$.

The probability mass function of a binomial random variable having parameters (n, p) is given by

$$p(i) = \binom{n}{i} p^i (1 - p)^{n-i} \quad i = 0, 1, \dots, n \tag{6.2}$$

The validity of Equation (6.2) may be verified by first noting that the probability of any particular sequence of n outcomes containing i successes and $n - i$ failures is, by the assumed independence of trials, $p^i (1 - p)^{n-i}$. Equation (6.2) then follows, since

there are $\binom{n}{i}$ different sequences of the n outcomes leading to i successes and

$n - i$ failures. This perhaps can most easily be seen by noting that there are $\binom{n}{i}$ different choices of the i trials that result in successes. For instance, if $n = 4, i = 2$,

then there are $\binom{4}{2} = 6$ ways in which the four trials can result in two successes,

namely, any of the outcomes $(s, s, f, f), (s, f, s, f), (s, f, f, s), (f, s, s, f), (f, s, f, s)$, and (f, f, s, s) , where the outcome (s, s, f, f) means, for instance, that the first two trials are successes and the last two failures. Since each of these outcomes has probability $p^2(1 - p)^2$ of occurring, the desired probability of two successes in the four trials

is $\binom{4}{2} p^2 (1 - p)^2$.

Note that, by the binomial theorem, the probabilities sum to 1; that is,

$$\sum_{i=0}^{\infty} p(i) = \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} = [p + (1-p)]^n = 1$$

**Example
6a**

Five fair coins are flipped. If the outcomes are assumed independent, find the probability mass function of the number of heads obtained.

Solution If we let X equal the number of heads (successes) that appear, then X is a binomial random variable with parameters $(n = 5, p = \frac{1}{2})$. Hence, by Equation (6.2),

$$\begin{aligned} P\{X = 0\} &= \binom{5}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{32} \\ P\{X = 1\} &= \binom{5}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{5}{32} \\ P\{X = 2\} &= \binom{5}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32} \\ P\{X = 3\} &= \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32} \\ P\{X = 4\} &= \binom{5}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{32} \\ P\{X = 5\} &= \binom{5}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32} \end{aligned}$$

**Example
6b**

It is known that screws produced by a certain company will be defective with probability .01, independently of one another. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?

Solution If X is the number of defective screws in a package, then X is a binomial random variable with parameters (10, .01). Hence, the probability that a package will have to be replaced is

$$\begin{aligned} 1 - P\{X = 0\} - P\{X = 1\} &= 1 - \binom{10}{0} (.01)^0 (.99)^{10} - \binom{10}{1} (.01)^1 (.99)^9 \\ &\approx .004 \end{aligned}$$

Thus, only .4 percent of the packages will have to be replaced.

**Example
6c**

The following gambling game, known as the wheel of fortune (or chuck-a-luck), is quite popular at many carnivals and gambling casinos: A player bets on one of the numbers 1 through 6. Three dice are then rolled, and if the number bet by the player appears i times, $i = 1, 2, 3$, then the player wins i units; if the number bet by the player does not appear on any of the dice, then the player loses 1 unit. Is this game fair to the player? (Actually, the game is played by spinning a wheel that comes to rest on a slot labeled by three of the numbers 1 through 6, but this variant is mathematically equivalent to the dice version.)

Solution If we assume that the dice are fair and act independently of one another, then the number of times that the number bet appears is a binomial random variable with parameters $(3, \frac{1}{6})$. Hence, letting X denote the player's winnings in the game, we have

$$P\{X = -1\} = \binom{3}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

$$P\{X = 1\} = \binom{3}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{75}{216}$$

$$P\{X = 2\} = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = \frac{15}{216}$$

$$P\{X = 3\} = \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = \frac{1}{216}$$

In order to determine whether or not this is a fair game for the player, let us calculate $E[X]$. From the preceding probabilities, we obtain

$$\begin{aligned} E[X] &= \frac{-125 + 75 + 30 + 3}{216} \\ &= \frac{-17}{216} \end{aligned}$$

Hence, in the long run, the player will lose 17 units per every 216 games he plays. ■

In the next example, we consider the simplest form of the theory of inheritance as developed by Gregor Mendel (1822–1884).

**Example
6d**

Suppose that a particular trait (such as eye color or left-handedness) of a person is classified on the basis of one pair of genes, and suppose also that d represents a dominant gene and r a recessive gene. Thus, a person with dd genes is purely dominant, one with rr is purely recessive, and one with rd is hybrid. The purely dominant and the hybrid individuals are alike in appearance. Children receive 1 gene from each parent. If, with respect to a particular trait, 2 hybrid parents have a total of 4 children, what is the probability that 3 of the 4 children have the outward appearance of the dominant gene?

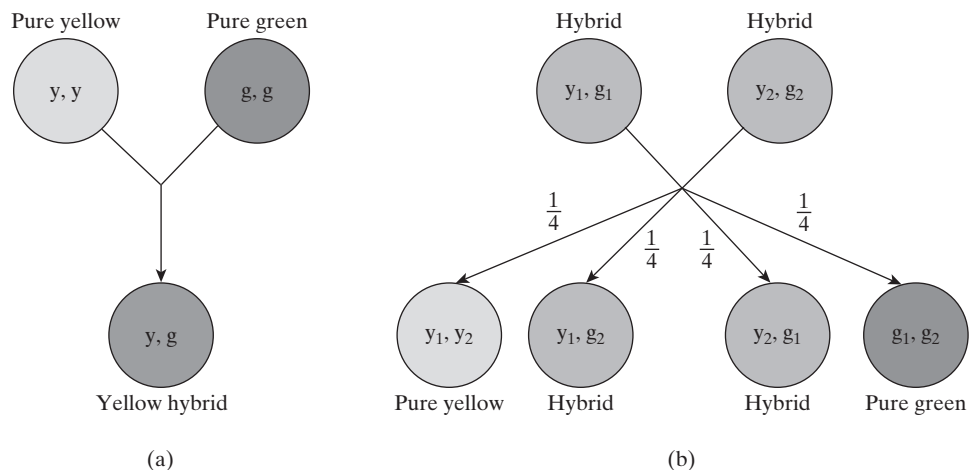


Figure 5 (a) Crossing pure yellow seeds with pure green seeds; (b) Crossing hybrid first-generation seeds.

The preceding Figure 5a and b shows what can happen when hybrid yellow (dominant) and green (recessive) seeds are crossed.

Solution If we assume that each child is equally likely to inherit either of 2 genes from each parent, the probabilities that the child of 2 hybrid parents will have dd , rr , and rd pairs of genes are, respectively, $\frac{1}{4}$, $\frac{1}{4}$, and $\frac{1}{2}$. Hence, since an offspring will have the outward appearance of the dominant gene if its gene pair is either dd or rd , it follows that the number of such children is binomially distributed with parameters $(4, \frac{3}{4})$. Thus, the desired probability is

$$\binom{4}{3} \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^1 = \frac{27}{64} \quad \blacksquare$$

**Example
6e**

Consider a jury trial in which it takes 8 of the 12 jurors to convict the defendant; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. If we assume that jurors act independently and that whether or not the defendant is guilty, each makes the right decision with probability θ , what is the probability that the jury renders a correct decision?

Solution The problem, as stated, is incapable of solution, for there is not yet enough information. For instance, if the defendant is innocent, the probability of the jury rendering a correct decision is

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1 - \theta)^{12-i}$$

whereas, if he is guilty, the probability of a correct decision is

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1 - \theta)^{12-i}$$

Therefore, if α represents the probability that the defendant is guilty, then, by conditioning on whether or not he is guilty, we obtain the probability that the jury renders a correct decision:

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1 - \theta)^{12-i} + (1 - \alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1 - \theta)^{12-i} \quad \blacksquare$$

**Example
6f**

A communication system consists of n components, each of which will, independently, function with probability p . The total system will be able to operate effectively if at least one-half of its components function.

- (a) For what values of p is a 5-component system more likely to operate effectively than a 3-component system?
- (b) In general, when is a $(2k + 1)$ -component system better than a $(2k - 1)$ -component system?

Solution (a) Because the number of functioning components is a binomial random variable with parameters (n, p) , it follows that the probability that a 5-component system will be effective is

$$\binom{5}{3} p^3 (1 - p)^2 + \binom{5}{4} p^4 (1 - p) + p^5$$

whereas the corresponding probability for a 3-component system is

$$\binom{3}{2} p^2 (1 - p) + p^3$$

Hence, the 5-component system is better if

$$10p^3(1 - p)^2 + 5p^4(1 - p) + p^5 > 3p^2(1 - p) + p^3$$

which reduces to

$$3(p - 1)^2(2p - 1) > 0$$

or

$$p > \frac{1}{2}$$

(b) In general, a system with $2k + 1$ components will be better than one with $2k - 1$ components if (and only if) $p > \frac{1}{2}$. To prove this, consider a system of $2k + 1$ components and let X denote the number of the first $2k - 1$ that function. Then

$$\begin{aligned} P_{2k+1}(\text{effective}) &= P\{X \geq k + 1\} + P\{X = k\}(1 - (1 - p)^2) \\ &\quad + P\{X = k - 1\}p^2 \end{aligned}$$

which follows because the $(2k + 1)$ -component system will be effective if either

- (i) $X \geq k + 1$;
- (ii) $X = k$ and at least one of the remaining 2 components function; or
- (iii) $X = k - 1$ and both of the next 2 components function.

Since

$$\begin{aligned} P_{2k-1}(\text{effective}) &= P\{X \geq k\} \\ &= P\{X = k\} + P\{X \geq k + 1\} \end{aligned}$$

we obtain

$$\begin{aligned} &P_{2k+1}(\text{effective}) - P_{2k-1}(\text{effective}) \\ &= P\{X = k - 1\}p^2 - (1 - p)^2 P\{X = k\} \\ &= \binom{2k - 1}{k - 1} p^{k-1} (1 - p)^k p^2 - (1 - p)^2 \binom{2k - 1}{k} p^k (1 - p)^{k-1} \\ &= \binom{2k - 1}{k} p^k (1 - p)^k [p - (1 - p)] \text{ since } \binom{2k - 1}{k - 1} = \binom{2k - 1}{k} \\ &> 0 \Leftrightarrow p > \frac{1}{2} \quad \blacksquare \end{aligned}$$

6.1 Properties of Binomial Random Variables

We will now examine the properties of a binomial random variable with parameters n and p . To begin, let us compute its expected value and variance. To begin, note that

$$\begin{aligned} E[X^k] &= \sum_{i=0}^n i^k \binom{n}{i} p^i (1 - p)^{n-i} \\ &= \sum_{i=1}^n i^k \binom{n}{i} p^i (1 - p)^{n-i} \end{aligned}$$

Using the identity

$$i \binom{n}{i} = n \binom{n-1}{i-1}$$

gives

$$\begin{aligned} E[X^k] &= np \sum_{i=1}^n i^{k-1} \binom{n-1}{i-1} p^{i-1} (1-p)^{n-i} \\ &= np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} \quad \text{by letting } j = i-1 \\ &= np E[(Y+1)^{k-1}] \end{aligned}$$

where Y is a binomial random variable with parameters $n-1, p$. Setting $k=1$ in the preceding equation yields

$$E[X] = np$$

That is, the expected number of successes that occur in n independent trials when each is a success with probability p is equal to np . Setting $k=2$ in the preceding equation and using the preceding formula for the expected value of a binomial random variable yields

$$\begin{aligned} E[X^2] &= np E[Y+1] \\ &= np[(n-1)p + 1] \end{aligned}$$

Since $E[X] = np$, we obtain

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= np[(n-1)p + 1] - (np)^2 \\ &= np(1-p) \end{aligned}$$

Summing up, we have shown the following:

If X is a binomial random variable with parameters n and p , then

$$\begin{aligned} E[X] &= np \\ \text{Var}(X) &= np(1-p) \end{aligned}$$

The following proposition details how the binomial probability mass function first increases and then decreases.

Proposition 6.1 If X is a binomial random variable with parameters (n, p) , where $0 < p < 1$, then as k goes from 0 to n , $P\{X = k\}$ first increases monotonically and then decreases monotonically, reaching its largest value when k is the largest integer less than or equal to $(n+1)p$.

Proof We prove the proposition by considering $P\{X = k\}/P\{X = k-1\}$ and determining for what values of k it is greater or less than 1. Now,

$$\begin{aligned} \frac{P\{X = k\}}{P\{X = k-1\}} &= \frac{\frac{n!}{(n-k)!k!} p^k (1-p)^{n-k}}{\frac{n!}{(n-k+1)!(k-1)!} p^{k-1} (1-p)^{n-k+1}} \\ &= \frac{(n-k+1)p}{k(1-p)} \end{aligned}$$

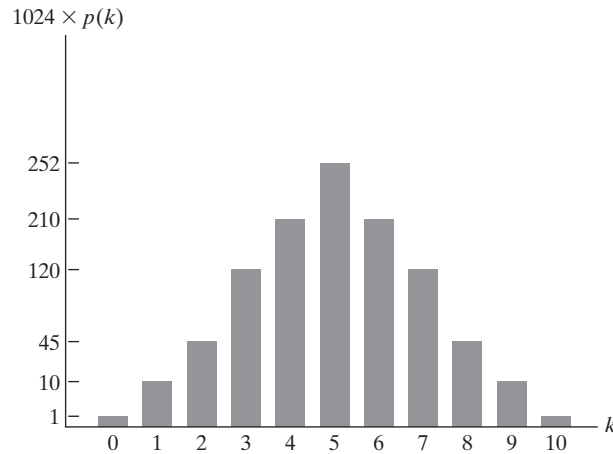


Figure 6 Graph of $p(k) = \binom{10}{k} \left(\frac{1}{2}\right)^{10}$.

Hence, $P\{X = k\} \geq P\{X = k - 1\}$ if and only if

$$(n - k + 1)p \geq k(1 - p)$$

or, equivalently, if and only if

$$k \leq (n + 1)p$$

and the proposition is proved. $\square$

As an illustration of Proposition 6.1, consider Figure 6, the graph of the probability mass function of a binomial random variable with parameters $(10, \frac{1}{2})$.

**Example
6g**

In a U.S. presidential election, the candidate who gains the maximum number of votes in a state is awarded the total number of electoral college votes allocated to that state. The number of electoral college votes of a given state is roughly proportional to the population of that state—that is, a state with population n has roughly nc electoral votes. (Actually, it is closer to $nc + 2$, as a state is given an electoral vote for each member it has in the House of Representatives, with the number of such representatives being roughly proportional to the population of the state, and one electoral college vote for each of its two senators.) Let us determine the average power of a citizen in a state of size n in a close presidential election, where, by *average power in a close election*, we mean that a voter in a state of size $n = 2k + 1$ will be decisive if the other $n - 1$ voters split their votes evenly between the two candidates. (We are assuming here that n is odd, but the case where n is even is quite similar.) Because the election is close, we shall suppose that each of the other $n - 1 = 2k$ voters acts independently and is equally likely to vote for either candidate. Hence, the probability that a voter in a state of size $n = 2k + 1$ will make a difference to the outcome is the same as the probability that $2k$ tosses of a fair coin land heads and tails an equal number of times. That is,

$$\begin{aligned} P\{\text{voter in state of size } 2k + 1 \text{ makes a difference}\} \\ &= \binom{2k}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^k \\ &= \frac{(2k)!}{k!k!2^{2k}} \end{aligned}$$

To approximate the preceding equality, we make use of Stirling's approximation, which says that for k large,

$$k! \sim k^{k+1/2} e^{-k} \sqrt{2\pi}$$

where we say that $a_k \sim b_k$ when the ratio a_k/b_k approaches 1 as k approaches ∞ . Hence, it follows that

$$\begin{aligned} &P\{\text{voter in state of size } 2k + 1 \text{ makes a difference}\} \\ &\sim \frac{(2k)^{2k+1/2} e^{-2k} \sqrt{2\pi}}{k^{2k+1} e^{-2k} (2\pi)^{2k}} = \frac{1}{\sqrt{k\pi}} \end{aligned}$$

Because such a voter (if he or she makes a difference) will affect nc electoral votes, the expected number of electoral votes a voter in a state of size n will affect—or the voter's average power—is given by

$$\begin{aligned} \text{average power} &= ncP\{\text{makes a difference}\} \\ &\sim \frac{nc}{\sqrt{n\pi/2}} \\ &= c\sqrt{2n/\pi} \end{aligned}$$

Thus, the average power of a voter in a state of size n is proportional to the square root of n , showing that in presidential elections, voters in large states have more power than do those in smaller states. ■

6.2 Computing the Binomial Distribution Function

Suppose that X is binomial with parameters (n, p) . The key to computing its distribution function

$$P\{X \leq i\} = \sum_{k=0}^i \binom{n}{k} p^k (1-p)^{n-k} \quad i = 0, 1, \dots, n$$

is to utilize the following relationship between $P\{X = k + 1\}$ and $P\{X = k\}$, which was established in the proof of Proposition 6.1:

$$P\{X = k + 1\} = \frac{p}{1-p} \frac{n-k}{k+1} P\{X = k\} \quad (6.3)$$

Example 6h

Let X be a binomial random variable with parameters $n = 6$, $p = .4$. Then, starting with $P\{X = 0\} = (.6)^6$ and recursively employing Equation (6.3), we obtain

$$P\{X = 0\} = (.6)^6 \approx .0467$$

$$P\{X = 1\} = \frac{4}{6} \frac{6}{1} P\{X = 0\} \approx .1866$$

$$P\{X = 2\} = \frac{4}{6} \frac{5}{2} P\{X = 1\} \approx .3110$$

$$P\{X = 3\} = \frac{4}{6} \frac{4}{3} P\{X = 2\} \approx .2765$$

$$P\{X = 4\} = \frac{4}{6} \frac{3}{4} P\{X = 3\} \approx .1382$$

$$P\{X = 5\} = \frac{4}{6} \frac{2}{5} P\{X = 4\} \approx .0369$$

$$P\{X = 6\} = \frac{4}{6} \frac{1}{6} P\{X = 5\} \approx .0041 \quad \blacksquare$$

A computer program that utilizes the recursion (6.3) to compute the binomial distribution function is easily written. To compute $P\{X \leq i\}$, the program should first compute $P\{X = i\}$ and then use the recursion to successively compute $P\{X = i - 1\}$, $P\{X = i - 2\}$, and so on.

Historical note

Independent trials having a common probability of success p were first studied by the Swiss mathematician Jacques Bernoulli (1654–1705). In his book *Ars Conjectandi* (*The Art of Conjecturing*), published by his nephew Nicholas eight years after his death in 1713, Bernoulli showed that if the number of such trials were large, then the proportion of them that were successes would be close to p with a probability near 1.

Jacques Bernoulli was from the first generation of the most famous mathematical family of all time. Altogether, there were between 8 and 12 Bernoullis, spread over three generations, who made fundamental contributions to probability, statistics, and mathematics. One difficulty in knowing their exact number is the fact that several had the same name. (For example, two of the sons of Jacques's brother Jean were named Jacques and Jean.) Another difficulty is that several of the Bernoullis were known by different names in different places. Our Jacques (sometimes written Jaques) was, for instance, also known as Jakob (sometimes written Jacob) and as James Bernoulli. But whatever their number, their influence and output were prodigious. Like the Bachs of music, the Bernoullis of mathematics were a family for the ages!

Example 6i

If X is a binomial random variable with parameters $n = 100$ and $p = .75$, find $P\{X = 70\}$ and $P\{X \leq 70\}$.

Solution A binomial calculator can be used to obtain the following solutions:

Figure 7

7 The Poisson Random Variable

A random variable X that takes on one of the values $0, 1, 2, \dots$ is said to be a *Poisson* random variable with parameter λ if, for some $\lambda > 0$,

$$p(i) = P\{X = i\} = e^{-\lambda} \frac{\lambda^i}{i!} \quad i = 0, 1, 2, \dots \quad (7.1)$$

Equation (7.1) defines a probability mass function, since

$$\sum_{i=0}^{\infty} p(i) = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1$$

The Poisson probability distribution was introduced by Siméon Denis Poisson in a book he wrote regarding the application of probability theory to lawsuits, criminal trials, and the like. This book, published in 1837, was entitled *Recherches sur la probabilité des jugements en matière criminelle et en matière civile (Investigations into the Probability of Verdicts in Criminal and Civil Matters)*.

The Poisson random variable has a tremendous range of applications in diverse areas because it may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small enough so that np is of moderate size. To see this, suppose that X is a binomial random variable with parameters (n, p) , and let $\lambda = np$. Then

$$\begin{aligned} P\{X = i\} &= \frac{n!}{(n-i)!i!} p^i (1-p)^{n-i} \\ &= \frac{n!}{(n-i)!i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i} \\ &= \frac{n(n-1) \cdots (n-i+1)}{n^i} \frac{\lambda^i}{i!} \left(1 - \frac{\lambda}{n}\right)^n \end{aligned}$$

Now, for n large and λ moderate,

$$\left(1 - \frac{\lambda}{n}\right)^n \approx e^{-\lambda} \quad \frac{n(n-1) \cdots (n-i+1)}{n^i} \approx 1 \quad \left(1 - \frac{\lambda}{n}\right)^i \approx 1$$

Hence, for n large and λ moderate,

$$P\{X = i\} \approx e^{-\lambda} \frac{\lambda^i}{i!}$$

In other words, if n independent trials, each of which results in a success with probability p , are performed, then when n is large and p is small enough to make np moderate, the number of successes occurring is approximately a Poisson random variable with parameter $\lambda = np$. This value λ (which will later be shown to equal the expected number of successes) will usually be determined empirically.

Some examples of random variables that generally obey the Poisson probability law [that is, they obey Equation (7.1)] are as follows:

1. The number of misprints on a page (or a group of pages) of a book
2. The number of people in a community who survive to age 100
3. The number of wrong telephone numbers that are dialed in a day
4. The number of packages of dog biscuits sold in a particular store each day
5. The number of customers entering a post office on a given day
6. The number of vacancies occurring during a year in the federal judicial system
7. The number of α -particles discharged in a fixed period of time from some radioactive material

Each of the preceding and numerous other random variables are approximately Poisson for the same reason—namely, because of the Poisson approximation to the binomial. For instance, we can suppose that there is a small probability p that each letter typed on a page will be misprinted. Hence, the number of misprints on a page will be approximately Poisson with $\lambda = np$, where n is the number of letters on a page. Similarly, we can suppose that each person in a community has some small

probability of reaching age 100. Also, each person entering a store may be thought of as having some small probability of buying a package of dog biscuits, and so on.

**Example
7a**

Suppose that the number of typographical errors on a single page of this book has a Poisson distribution with parameter $\lambda = \frac{1}{2}$. Calculate the probability that there is at least one error on this page.

Solution Letting X denote the number of errors on this page, we have

$$P\{X \geq 1\} = 1 - P\{X = 0\} = 1 - e^{-1/2} \approx .393 \quad \blacksquare$$

**Example
7b**

Suppose that the probability that an item produced by a certain machine will be defective is .1. Find the probability that a sample of 10 items will contain at most 1 defective item.

Solution The desired probability is $\binom{10}{0} (.1)^0 (.9)^{10} + \binom{10}{1} (.1)^1 (.9)^9 = .7361$, whereas the Poisson approximation yields the value $e^{-1} + e^{-1} \approx .7358$. $\blacksquare$

**Example
7c**

Consider an experiment that consists of counting the number of α particles given off in a 1-second interval by 1 gram of radioactive material. If we know from past experience that on the average, 3.2 such α particles are given off, what is a good approximation to the probability that no more than 2 α particles will appear?

Solution If we think of the gram of radioactive material as consisting of a large number n of atoms, each of which has probability of $3.2/n$ of disintegrating and sending off an α particle during the second considered, then we see that to a very close approximation, the number of α particles given off will be a Poisson random variable with parameter $\lambda = 3.2$. Hence, the desired probability is

$$\begin{aligned} P\{X \leq 2\} &= e^{-3.2} + 3.2e^{-3.2} + \frac{(3.2)^2}{2}e^{-3.2} \\ &\approx .3799 \end{aligned} \quad \blacksquare$$

Compute the expected value and variance of the Poisson random variable with parameter λ ; this random variable approximates a binomial random variable with parameters n and p when n is large, p is small, and $\lambda = np$. Since such a binomial random variable has expected value $np = \lambda$ and variance $np(1 - p) = \lambda(1 - p) \approx \lambda$ (since p is small), it would seem that both the expected value and the variance of a Poisson random variable would equal its parameter λ . We now verify this result:

$$\begin{aligned} E[X] &= \sum_{i=0}^{\infty} \frac{ie^{-\lambda}\lambda^i}{i!} \\ &= \lambda \sum_{i=1}^{\infty} \frac{e^{-\lambda}\lambda^{i-1}}{(i-1)!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \quad \text{by letting } j = i - 1 \\ &= \lambda \quad \text{since } \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = e^{\lambda} \end{aligned}$$

Thus, the expected value of a Poisson random variable X is indeed equal to its parameter λ . To determine its variance, we first compute $E[X^2]$:

$$\begin{aligned}
 E[X^2] &= \sum_{i=0}^{\infty} \frac{i^2 e^{-\lambda} \lambda^i}{i!} \\
 &= \lambda \sum_{i=1}^{\infty} \frac{i e^{-\lambda} \lambda^{i-1}}{(i-1)!} \\
 &= \lambda \sum_{j=0}^{\infty} \frac{(j+1) e^{-\lambda} \lambda^j}{j!} \quad \text{by letting } j = i - 1 \\
 &= \lambda \left[\sum_{j=0}^{\infty} \frac{j e^{-\lambda} \lambda^j}{j!} + \sum_{j=0}^{\infty} \frac{e^{-\lambda} \lambda^j}{j!} \right] \\
 &= \lambda(\lambda + 1)
 \end{aligned}$$

where the final equality follows because the first sum is the expected value of a Poisson random variable with parameter λ and the second is the sum of the probabilities of this random variable. Therefore, since we have shown that $E[X] = \lambda$, we obtain

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - (E[X])^2 \\
 &= \lambda
 \end{aligned}$$

Hence, the expected value and variance of a Poisson random variable are both equal to its parameter λ .

We have shown that the Poisson distribution with parameter np is a very good approximation to the distribution of the number of successes in n independent trials when each trial has probability p of being a success, provided that n is large and p small. In fact, it remains a good approximation even when the trials are not independent, provided that their dependence is weak. For instance, consider a matching problem in which n men randomly select hats from a set consisting of one hat from each person. From the point of view of the number of men who select their own hat, we may regard the random selection as the result of n trials where we say that trial i is a success if person i selects his own hat, $i = 1, \dots, n$. Defining the events $E_i, i = 1, \dots, n$, by

$$E_i = \{\text{trial } i \text{ is a success}\}$$

it is easy to see that

$$P\{E_i\} = \frac{1}{n} \quad \text{and} \quad P\{E_i|E_j\} = \frac{1}{n-1}, \quad j \neq i$$

Thus, we see that although the events $E_i, i = 1, \dots, n$ are not independent, their dependence, for large n , appears to be weak. Because of this, it seems reasonable to expect that the number of successes will approximately have a Poisson distribution with parameter $n \times 1/n = 1$.

For a second illustration of the strength of the Poisson approximation when the trials are weakly dependent, let us consider an example, suppose that each of n people is equally likely to have any of the 365 days of the year as his or her birthday, and the problem is to determine the probability that a set of n independent people all have different birthdays.

We can approximate the probability by using the Poisson approximation as follows: Imagine that we have a trial for each of the $\binom{n}{2}$ pairs of individuals i and $j, i \neq j$, and say that trial i, j is a success if persons i and j have the same birthday. If we let E_{ij} denote the event that trial i, j is a success, then, whereas the $\binom{n}{2}$ events $E_{ij}, 1 \leq i < j \leq n$, are not independent (see Theoretical Exercise 21), their dependence appears to be rather weak. (Indeed, these events are even *pairwise independent*, in that any 2 of the events E_{ij} and E_{kl} are independent—again, see Theoretical Exercise 21). Since $P(E_{ij}) = 1/365$, it is reasonable to suppose that the number of successes should approximately have a Poisson distribution with mean $\binom{n}{2} / 365 = n(n - 1)/730$. Therefore,

$$\begin{aligned} P\{\text{no 2 people have the same birthday}\} &= P\{0 \text{ successes}\} \\ &\approx \exp\left\{\frac{-n(n - 1)}{730}\right\} \end{aligned}$$

To determine the smallest integer n for which this probability is less than $\frac{1}{2}$, note that

$$\exp\left\{\frac{-n(n - 1)}{730}\right\} \leq \frac{1}{2}$$

is equivalent to

$$\exp\left\{\frac{n(n - 1)}{730}\right\} \geq 2$$

Taking logarithms of both sides, we obtain

$$\begin{aligned} n(n - 1) &\geq 730 \log 2 \\ &\approx 505.997 \end{aligned}$$

which yields the solution $n = 23$.

Suppose now that we wanted the probability that among the n people, no 3 of them have their birthday on the same day. Whereas this now becomes a difficult combinatorial problem, it is a simple matter to obtain a good approximation. To begin, imagine that we have a trial for each of the $\binom{n}{3}$ triplets i, j, k , where $1 \leq i < j < k \leq n$, and call the i, j, k trial a success if persons i, j , and k all have their birthday on the same day. As before, we can then conclude that the number of successes is approximately a Poisson random variable with parameter

$$\begin{aligned} \binom{n}{3} P\{i, j, k \text{ have the same birthday}\} &= \binom{n}{3} \left(\frac{1}{365}\right)^2 \\ &= \frac{n(n - 1)(n - 2)}{6 \times (365)^2} \end{aligned}$$

Hence,

$$P\{\text{no 3 have the same birthday}\} \approx \exp\left\{\frac{-n(n - 1)(n - 2)}{799350}\right\}$$

This probability will be less than $\frac{1}{2}$ when n is such that

$$n(n-1)(n-2) \geq 799350 \log 2 \approx 554067.1$$

which is equivalent to $n \geq 84$. Thus, the approximate probability that at least 3 people in a group of size 84 or larger will have the same birthday exceeds $\frac{1}{2}$.

For the number of events to occur to approximately have a Poisson distribution, it is not essential that all the events have the same probability of occurrence, but only that all of these probabilities be small. The following is referred to as the *Poisson paradigm*.

Poisson Paradigm. Consider n events, with p_i equal to the probability that event i occurs, $i = 1, \dots, n$. If all the p_i are “small” and the trials are either independent or at most “weakly dependent,” then the number of these events that occur approximately has a Poisson distribution with mean $\sum_{i=1}^n p_i$.

Our next example not only makes use of the Poisson paradigm, but also illustrates a variety of the techniques we have studied so far.

Example 7d

Length of the longest run

A coin is flipped n times. Assuming that the flips are independent, with each one coming up heads with probability p , what is the probability that there is a string of k consecutive heads?

Solution We will first use the Poisson paradigm to approximate this probability. Now, if for $i = 1, \dots, n - k + 1$, we let H_i denote the event that flips $i, i + 1, \dots, i + k - 1$ all land on heads, then the desired probability is that at least one of the events H_i occur. Because H_i is the event that starting with flip i , the next k flips all land on heads, it follows that $P(H_i) = p^k$. Thus, when p^k is small, we might think that the number of the H_i that occur should have an approximate Poisson distribution. However, such is not the case, because, although the events all have small probabilities, some of their dependencies are too great for the Poisson distribution to be a good approximation. For instance, because the conditional probability that flips $2, \dots, k + 1$ are all heads given that flips $1, \dots, k$ are all heads is equal to the probability that flip $k + 1$ is a head, it follows that

$$P(H_2|H_1) = p$$

which is far greater than the unconditional probability of H_2 .

The trick that enables us to use a Poisson approximation is to note that there will be a string of k consecutive heads either if there is such a string that is immediately followed by a tail or if the final k flips all land on heads. Consequently, for $i = 1, \dots, n - k$, let E_i be the event that flips $i, \dots, i + k - 1$ are all heads and flip $i + k$ is a tail; also, let E_{n-k+1} be the event that flips $n - k + 1, \dots, n$ are all heads. Note that

$$\begin{aligned} P(E_i) &= p^k(1 - p), \quad i \leq n - k \\ P(E_{n-k+1}) &= p^k \end{aligned}$$

Thus, when p^k is small, each of the events E_i has a small probability of occurring. Moreover, for $i \neq j$, if the events E_i and E_j refer to nonoverlapping sequences of flips, then $P(E_i|E_j) = P(E_i)$; if they refer to overlapping sequences, then $P(E_i|E_j) = 0$. Hence, in both cases, the conditional probabilities are close to the unconditional

ones, indicating that N , the number of the events E_i that occur, should have an approximate Poisson distribution with mean

$$E[N] = \sum_{i=1}^{n-k+1} P(E_i) = (n - k)p^k(1 - p) + p^k$$

Because there will not be a run of k heads if (and only if) $N = 0$, thus the preceding gives

$$P(\text{no head strings of length } k) = P(N = 0) \approx \exp\{-(n - k)p^k(1 - p) - p^k\}$$

If we let L_n denote the largest number of consecutive heads in the n flips, then, because L_n will be less than k if (and only if) there are no head strings of length k , the preceding equation can be written as

$$P\{L_n < k\} \approx \exp\{-(n - k)p^k(1 - p) - p^k\}$$

Now, let us suppose that the coin being flipped is fair; that is, suppose that $p = 1/2$. Then the preceding gives

$$P\{L_n < k\} \approx \exp\left\{-\frac{n - k + 2}{2^{k+1}}\right\} \approx \exp\left\{-\frac{n}{2^{k+1}}\right\}$$

where the final approximation supposes that $e^{\frac{k-2}{2^{k+1}}} \approx 1$ (that is, that $\frac{k-2}{2^{k+1}} \approx 0$). Let $j = \log_2 n$, and assume that j is an integer. For $k = j + i$,

$$\frac{n}{2^{k+1}} = \frac{n}{2^{j+i+1}} = \frac{1}{2^{i+1}}$$

Consequently,

$$P\{L_n < j + i\} \approx \exp\{-(1/2)^{i+1}\}$$

which implies that

$$\begin{aligned} P\{L_n = j + i\} &= P\{L_n < j + i + 1\} - P\{L_n < j + i\} \\ &\approx \exp\{-(1/2)^{i+2}\} - \exp\{-(1/2)^{i+1}\} \end{aligned}$$

For instance,

$$\begin{aligned} P\{L_n < j - 3\} &\approx e^{-4} \approx .0183 \\ P\{L_n = j - 3\} &\approx e^{-2} - e^{-4} \approx .1170 \\ P\{L_n = j - 2\} &\approx e^{-1} - e^{-2} \approx .2325 \\ P\{L_n = j - 1\} &\approx e^{-1/2} - e^{-1} \approx .2387 \\ P\{L_n = j\} &\approx e^{-1/4} - e^{-1/2} \approx .1723 \\ P\{L_n = j + 1\} &\approx e^{-1/8} - e^{-1/4} \approx .1037 \\ P\{L_n = j + 2\} &\approx e^{-1/16} - e^{-1/8} \approx .0569 \\ P\{L_n = j + 3\} &\approx e^{-1/32} - e^{-1/16} \approx .0298 \\ P\{L_n \geq j + 4\} &\approx 1 - e^{-1/32} \approx .0308 \end{aligned}$$

Thus, we observe the rather interesting fact that no matter how large n is, the length of the longest run of heads in a sequence of n flips of a fair coin will be within 2 of $\log_2(n) - 1$ with a probability approximately equal to .86.

We now derive an exact expression for the probability that there is a string of k consecutive heads when a coin that lands on heads with probability p is flipped n times. With the events $E_i, i = 1, \dots, n - k + 1$, as defined earlier, and with L_n denoting, as before, the length of the longest run of heads,

$$P(L_n \geq k) = P(\text{there is a string of } k \text{ consecutive heads}) = P(\cup_{i=1}^{n-k+1} E_i)$$

The inclusion–exclusion identity for the probability of a union can be written as

$$P(\cup_{i=1}^{n-k+1} E_i) = \sum_{r=1}^{n-k+1} (-1)^{r+1} \sum_{i_1 < \dots < i_r} P(E_{i_1} \cdots E_{i_r})$$

Let S_i denote the set of flip numbers to which the event E_i refers. (So, for instance, $S_1 = \{1, \dots, k + 1\}$.) Now, consider one of the r -way intersection probabilities that does not include the event E_{n-k+1} . That is, consider $P(E_{i_1} \cdots E_{i_r})$ where $i_1 < \dots < i_r < n - k + 1$. On the one hand, if there is any overlap in the sets $S_{i_1}, \dots, S_{i_r}$ then this probability is 0. On the other hand, if there is no overlap, then the events $E_{i_1}, \dots, E_{i_r}$ are independent. Therefore,

$$P(E_{i_1} \cdots E_{i_r}) = \begin{cases} 0, & \text{if there is any overlap in } S_{i_1}, \dots, S_{i_r} \\ p^{rk}(1 - p)^r, & \text{if there is no overlap} \end{cases}$$

We must now determine the number of different choices of $i_1 < \dots < i_r < n - k + 1$ for which there is no overlap in the sets $S_{i_1}, \dots, S_{i_r}$. To do so, note first that each of the $S_{i_j}, j = 1, \dots, r$, refer to $k + 1$ flips, so, without any overlap, they together refer to $r(k + 1)$ flips. Now consider any permutation of r identical letters a and of $n - r(k + 1)$ identical letters b . Interpret the number of b 's before the first a as the number of flips before S_{i_1} , the number of b 's between the first and second a as the number of flips between S_{i_1} and S_{i_2} , and so on, with the number of b 's after the final a representing the number of flips after S_{i_r} . Because there are $\binom{n-rk}{r}$ permutations of r letters a and of $n - r(k + 1)$ letters b , with every such permutation corresponding (in a one-to-one fashion) to a different nonoverlapping choice, it follows that

$$\sum_{i_1 < \dots < i_r < n-k+1} P(E_{i_1} \cdots E_{i_r}) = \binom{n-rk}{r} p^{rk}(1 - p)^r$$

We must now consider r -way intersection probabilities of the form

$$P(E_{i_1} \cdots E_{i_{r-1}} E_{n-k+1}),$$

where $i_1 < \dots < i_{r-1} < n - k + 1$. Now, this probability will equal 0 if there is any overlap in $S_{i_1}, \dots, S_{i_{r-1}}, S_{n-k}$; if there is no overlap, then the events of the intersection will be independent, so

$$P(E_{i_1} \cdots E_{i_{r-1}} E_{n-k+1}) = [p^k(1 - p)]^{r-1} p^k = p^{kr}(1 - p)^{r-1}$$

By a similar argument as before, the number of nonoverlapping sets $S_{i_1}, \dots, S_{i_{r-1}}, S_{n-k}$ will equal the number of permutations of $r - 1$ letters a (one for each of the sets $S_{i_1}, \dots, S_{i_{r-1}}$) and of $n - (r - 1)(k + 1) - k = n - rk - (r - 1)$ letters b (one for each of the trials that are not part of any of $S_{i_1}, \dots, S_{i_{r-1}}, S_{n-k+1}$). Since there are $\binom{n-rk}{r-1}$ permutations of $r - 1$ letters a and of $n - rk - (r - 1)$ letters b , we have

$$\sum_{i_1 < \dots < i_{r-1} < n-k+1} P(E_{i_1} \dots E_{i_{r-1}} E_{n-k+1}) = \binom{n-rk}{r-1} p^{kr} (1-p)^{r-1}$$

Putting it all together yields the exact expression, namely,

$$P(L_n \geq k) = \sum_{r=1}^{n-k+1} (-1)^{r+1} \left[\binom{n-rk}{r} + \frac{1}{p} \binom{n-rk}{r-1} \right] p^{kr} (1-p)^r$$

where we utilize the convention that $\binom{m}{j} = 0$ if $m < j$.

From a computational point of view, a more efficient method for computing the desired probability than the use of the preceding identity is to derive a set of recursive equations. To do so, let A_n be the event that there is a string of k consecutive heads in a sequence of n flips of a fair coin, and let $P_n = P(A_n)$. We will derive a set of recursive equations for P_n by conditioning on when the first tail appears. For $j = 1, \dots, k$, let F_j be the event that the first tail appears on flip j , and let H be the event that the first k flips are all heads. Because the events $F_1, \dots, F_k, H$ are mutually exclusive and exhaustive (that is, exactly one of these events must occur), we have

$$P(A_n) = \sum_{j=1}^k P(A_n | F_j) P(F_j) + P(A_n | H) P(H)$$

Now, given that the first tail appears on flip j , where $j < k$, it follows that those j flips are wasted as far as obtaining a string of k heads in a row; thus, the conditional probability of this event is the probability that such a string will occur among the remaining $n - j$ flips. Therefore,

$$P(A_n | F_j) = P_{n-j}$$

Because $P(A_n | H) = 1$, the preceding equation gives

$$\begin{aligned} P_n &= P(A_n) \\ &= \sum_{j=1}^k P_{n-j} P(F_j) + P(H) \\ &= \sum_{j=1}^k P_{n-j} p^{j-1} (1-p) + p^k \end{aligned}$$

Starting with $P_j = 0$, $j < k$, and $P_k = p^k$, we can use the latter formula to recursively compute P_{k+1} , P_{k+2} , and so on, up to P_n . For instance, suppose we want the probability that there is a run of 2 consecutive heads when a fair coin is flipped 4 times. Then, with $k = 2$, we have $P_1 = 0$, $P_2 = (1/2)^2$. Because, when $p = 1/2$, the recursion becomes

$$P_n = \sum_{j=1}^k P_{n-j} (1/2)^j + (1/2)^k$$

we obtain

$$P_3 = P_2(1/2) + P_1(1/2)^2 + (1/2)^2 = 3/8$$

and

$$P_4 = P_3(1/2) + P_2(1/2)^2 + (1/2)^2 = 1/2$$

which is clearly true because there are 8 outcomes that result in a string of 2 consecutive heads: $hhhh$, $hhht$, $hhth$, $hthh$, $thhh$, $hhtt$, $thht$, and $tthh$. Each of these outcomes occurs with probability $1/16$. ■

Another use of the Poisson probability distribution arises in situations where “events” occur at certain points in time. One example is to designate the occurrence of an earthquake as an event; another possibility would be for events to correspond to people entering a particular establishment (bank, post office, gas station, and so on); and a third possibility is for an event to occur whenever a war starts. Let us suppose that events are indeed occurring at certain (random) points of time, and let us assume that for some positive constant λ , the following assumptions hold true:

1. The probability that exactly 1 event occurs in a given interval of length h is equal to $\lambda h + o(h)$, where $o(h)$ stands for any function $f(h)$ for which $\lim_{h \rightarrow 0} f(h)/h = 0$. [For instance, $f(h) = h^2$ is $o(h)$, whereas $f(h) = h$ is not.]
2. The probability that 2 or more events occur in an interval of length h is equal to $o(h)$.
3. For any integers n , $j_1, j_2, \dots, j_n$ and any set of n nonoverlapping intervals, if we define E_i to be the event that exactly j_i of the events under consideration occur in the i th of these intervals, then events $E_1, E_2, \dots, E_n$ are independent.

Loosely put, assumptions 1 and 2 state that for small values of h , the probability that exactly 1 event occurs in an interval of size h equals λh plus something that is small compared with h , whereas the probability that 2 or more events occur is small compared with h . Assumption 3 states that whatever occurs in one interval has no (probability) effect on what will occur in other, nonoverlapping intervals.

We now show that under assumptions 1, 2, and 3, the number of events occurring in any interval of length t is a Poisson random variable with parameter λt . To be precise, let us call the interval $[0, t]$ and denote the number of events occurring in that interval by $N(t)$. To obtain an expression for $P\{N(t) = k\}$, we start by breaking the interval $[0, t]$ into n nonoverlapping subintervals, each of length t/n (Figure 8).



Figure 8

Now,

$$\begin{aligned}
 P\{N(t) = k\} &= P\{k \text{ of the } n \text{ subintervals contain exactly 1 event} \\
 &\quad \text{and the other } n - k \text{ contain 0 events}\} \\
 &+ P\{N(t) = k \text{ and at least 1 subinterval contains} \\
 &\quad \text{2 or more events}\}
 \end{aligned} \tag{7.2}$$

The preceding equation holds because the event on the left side of Equation (7.2), that is, $\{N(t) = k\}$, is clearly equal to the union of the two mutually exclusive events on the right side of the equation. Letting A and B denote the two mutually exclusive events on the right side of Equation (7.2), we have

$$\begin{aligned}
 P(B) &\leq P\{\text{at least one subinterval contains 2 or more events}\} \\
 &= P\left(\bigcup_{i=1}^n \{\text{ith subinterval contains 2 or more events}\}\right) \\
 &\leq \sum_{i=1}^n P\{\text{ith subinterval contains 2 or more events}\} \quad \text{by Boole's inequality} \\
 &= \sum_{i=1}^n o\left(\frac{t}{n}\right) \quad \text{by assumption 2} \\
 &= no\left(\frac{t}{n}\right) \\
 &= t \left[\frac{o(t/n)}{t/n} \right]
 \end{aligned}$$

Now, in addition, for any $t, t/n \rightarrow 0$ as $n \rightarrow \infty$, so $o(t/n)/(t/n) \rightarrow 0$ as $n \rightarrow \infty$, by the definition of $o(h)$. Hence,

$$P(B) \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad (7.3)$$

Moreover, since assumptions 1 and 2 imply that[†]

$$\begin{aligned}
 P\{0 \text{ events occur in an interval of length } h\} \\
 = 1 - [\lambda h + o(h) + o(h)] = 1 - \lambda h - o(h)
 \end{aligned}$$

we see from the independence assumption (number 3) that

$$\begin{aligned}
 P(A) &= P\{k \text{ of the subintervals contain exactly 1 event and the other} \\
 &\quad n - k \text{ contain 0 events}\} \\
 &= \binom{n}{k} \left[\frac{\lambda t}{n} + o\left(\frac{t}{n}\right) \right]^k \left[1 - \left(\frac{\lambda t}{n}\right) - o\left(\frac{t}{n}\right) \right]^{n-k}
 \end{aligned}$$

However, since

$$n \left[\frac{\lambda t}{n} + o\left(\frac{t}{n}\right) \right] = \lambda t + t \left[\frac{o(t/n)}{t/n} \right] \rightarrow \lambda t \quad \text{as } n \rightarrow \infty$$

it follows, by the same argument that verified the Poisson approximation to the binomial, that

$$P(A) \rightarrow e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad \text{as } n \rightarrow \infty \quad (7.4)$$

Thus, from Equations (7.2), (7.3), and (7.4), by letting $n \rightarrow \infty$, we obtain

$$P\{N(t) = k\} = e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad k = 0, 1, \dots \quad (7.5)$$

Hence, if assumptions 1, 2, and 3 are satisfied, then the number of events occurring in any fixed interval of length t is a Poisson random variable with mean λt , and we say that the events occur in accordance with a Poisson process having rate λ . The value λ , which can be shown to equal the rate per unit time at which events occur, is a constant that must be empirically determined.

[†]The sum of two functions, both of which are $o(h)$, is also $o(h)$. This is so because if $\lim_{h \rightarrow 0} f(h)/h = \lim_{h \rightarrow 0} g(h)/h = 0$, then $\lim_{h \rightarrow 0} [f(h) + g(h)]/h = 0$.

The preceding discussion explains why a Poisson random variable is usually a good approximation for such diverse phenomena as the following:

1. The number of earthquakes occurring during some fixed time span
2. The number of wars per year
3. The number of electrons emitted from a heated cathode during a fixed time period
4. The number of deaths, in a given period of time, of the policyholders of a life insurance company

**Example
7e**

Suppose that earthquakes occur in the western portion of the United States in accordance with assumptions 1, 2, and 3, with $\lambda = 2$ and with 1 week as the unit of time. (That is, earthquakes occur in accordance with the three assumptions at a rate of 2 per week.)

- (a) Find the probability that at least 3 earthquakes occur during the next 2 weeks.
- (b) Find the probability distribution of the time, starting from now, until the next earthquake.

Solution (a) From Equation (7.5), we have

$$\begin{aligned} P\{N(2) \geq 3\} &= 1 - P\{N(2) = 0\} - P\{N(2) = 1\} - P\{N(2) = 2\} \\ &= 1 - e^{-4} - 4e^{-4} - \frac{4^2}{2}e^{-4} \\ &= 1 - 13e^{-4} \end{aligned}$$

(b) Let X denote the amount of time (in weeks) until the next earthquake. Because X will be greater than t if and only if no events occur within the next t units of time, we have, from Equation (7.5),

$$P\{X > t\} = P\{N(t) = 0\} = e^{-\lambda t}$$

so the probability distribution function F of the random variable X is given by

$$\begin{aligned} F(t) = P\{X \leq t\} &= 1 - P\{X > t\} = 1 - e^{-\lambda t} \\ &= 1 - e^{-2t} \end{aligned}$$

■

7.1 Computing the Poisson Distribution Function

If X is Poisson with parameter λ , then

$$\frac{P\{X = i + 1\}}{P\{X = i\}} = \frac{e^{-\lambda} \lambda^{i+1} / (i + 1)!}{e^{-\lambda} \lambda^i / i!} = \frac{\lambda}{i + 1} \quad (7.6)$$

Starting with $P\{X = 0\} = e^{-\lambda}$, we can use (7.6) to compute successively

$$\begin{aligned} P\{X = 1\} &= \lambda P\{X = 0\} \\ P\{X = 2\} &= \frac{\lambda}{2} P\{X = 1\} \\ &\vdots \\ P\{X = i + 1\} &= \frac{\lambda}{i + 1} P\{X = i\} \end{aligned}$$

We can use a module to compute the Poisson probabilities for Equation (7.6).

**Example
7f**

- (a) Determine $P\{X \leq 90\}$ when X is Poisson with mean 100.
 (b) Determine $P\{Y \leq 1075\}$ when Y is Poisson with mean 1000.

Solution Using the Poisson calculator of StatCrunch yields the solutions:

- (a) $P\{X \leq 90\} = .17138$
 (b) $P\{Y \leq 1075\} = .99095$ ■

8 Other Discrete Probability Distributions

8.1 The Geometric Random Variable

Suppose that independent trials, each having a probability $p, 0 < p < 1$, of being a success, are performed until a success occurs. If we let X equal the number of trials required, then

$$P\{X = n\} = (1 - p)^{n-1}p \quad n = 1, 2, \dots \quad (8.1)$$

Equation (8.1) follows because, in order for X to equal n , it is necessary and sufficient that the first $n - 1$ trials are failures and the n th trial is a success. Equation (8.1) then follows, since the outcomes of the successive trials are assumed to be independent.

Since

$$\sum_{n=1}^{\infty} P\{X = n\} = p \sum_{n=1}^{\infty} (1 - p)^{n-1} = \frac{p}{1 - (1 - p)} = 1$$

it follows that with probability 1, a success will eventually occur. Any random variable X whose probability mass function is given by Equation (8.1) is said to be a *geometric* random variable with parameter p .

**Example
8a**

An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn, what is the probability that

- (a) exactly n draws are needed?
 (b) at least k draws are needed?

Solution If we let X denote the number of draws needed to select a black ball, then X satisfies Equation (8.1) with $p = M/(M + N)$. Hence,

(a)

$$P\{X = n\} = \left(\frac{N}{M + N}\right)^{n-1} \frac{M}{M + N} = \frac{MN^{n-1}}{(M + N)^n}$$

(b)

$$\begin{aligned} P\{X \geq k\} &= \frac{M}{M + N} \sum_{n=k}^{\infty} \left(\frac{N}{M + N}\right)^{n-1} \\ &= \left(\frac{M}{M + N}\right) \left(\frac{N}{M + N}\right)^{k-1} \left/ \left[1 - \frac{N}{M + N}\right] \right. \\ &= \left(\frac{N}{M + N}\right)^{k-1} \end{aligned}$$

Of course, part (b) could have been obtained directly, since the probability that at least k trials are necessary to obtain a success is equal to the probability that the first $k - 1$ trials are all failures. That is, for a geometric random variable,

$$P\{X \geq k\} = (1 - p)^{k-1}$$

**Example
8b**

Find the expected value of a geometric random variable.

Solution With $q = 1 - p$, we have

$$\begin{aligned} E[X] &= \sum_{i=1}^{\infty} i q^{i-1} p \\ &= \sum_{i=1}^{\infty} (i - 1 + 1) q^{i-1} p \\ &= \sum_{i=1}^{\infty} (i - 1) q^{i-1} p + \sum_{i=1}^{\infty} q^{i-1} p \\ &= \sum_{j=0}^{\infty} j q^j p + 1 \\ &= q \sum_{j=1}^{\infty} j q^{j-1} p + 1 \\ &= q E[X] + 1 \end{aligned}$$

Hence,

$$p E[X] = 1$$

yielding the result

$$E[X] = \frac{1}{p}$$

In other words, if independent trials having a common probability p of being successful are performed until the first success occurs, then the expected number of required trials equals $1/p$. For instance, the expected number of rolls of a fair die that it takes to obtain the value 1 is 6. ■

**Example
8c**

Find the variance of a geometric random variable.

Solution To determine $\text{Var}(X)$, let us first compute $E[X^2]$. With $q = 1 - p$, we have

$$\begin{aligned} E[X^2] &= \sum_{i=1}^{\infty} i^2 q^{i-1} p \\ &= \sum_{i=1}^{\infty} (i - 1 + 1)^2 q^{i-1} p \\ &= \sum_{i=1}^{\infty} (i - 1)^2 q^{i-1} p + \sum_{i=1}^{\infty} 2(i - 1) q^{i-1} p + \sum_{i=1}^{\infty} q^{i-1} p \\ &= \sum_{j=0}^{\infty} j^2 q^j p + 2 \sum_{j=1}^{\infty} j q^j p + 1 \\ &= q E[X^2] + 2q E[X] + 1 \end{aligned}$$

Using $E[X] = 1/p$, the equation for $E[X^2]$ yields

$$pE[X^2] = \frac{2q}{p} + 1$$

Hence,

$$E[X^2] = \frac{2q + p}{p^2} = \frac{q + 1}{p^2}$$

giving the result

$$\text{Var}(X) = \frac{q + 1}{p^2} - \frac{1}{p^2} = \frac{q}{p^2} = \frac{1 - p}{p^2} \quad \blacksquare$$

8.2 The Negative Binomial Random Variable

Suppose that independent trials, each having probability $p, 0 < p < 1$, of being a success are performed until a total of r successes is accumulated. If we let X equal the number of trials required, then

$$P\{X = n\} = \binom{n-1}{r-1} p^r (1-p)^{n-r} \quad n = r, r+1, \dots \quad (8.2)$$

Equation (8.2) follows because, in order for the r th success to occur at the n th trial, there must be $r-1$ successes in the first $n-1$ trials and the n th trial must be a success. The probability of the first event is

$$\binom{n-1}{r-1} p^{r-1} (1-p)^{n-r}$$

and the probability of the second is p ; thus, by independence, Equation (8.2) is established. To verify that a total of r successes must eventually be accumulated, either we can prove analytically that

$$\sum_{n=r}^{\infty} P\{X = n\} = \sum_{n=r}^{\infty} \binom{n-1}{r-1} p^r (1-p)^{n-r} = 1 \quad (8.3)$$

or we can give a probabilistic argument as follows: The number of trials required to obtain r successes can be expressed as $Y_1 + Y_2 + \dots + Y_r$, where Y_1 equals the number of trials required for the first success, Y_2 the number of additional trials after the first success until the second success occurs, Y_3 the number of additional trials until the third success, and so on. Because the trials are independent and all have the same probability of success, it follows that $Y_1, Y_2, \dots, Y_r$ are all geometric random variables. Hence, each is finite with probability 1, so $\sum_{i=1}^r Y_i$ must also be finite, establishing Equation (8.3).

Any random variable X whose probability mass function is given by Equation (8.2) is said to be a *negative binomial* random variable with parameters (r, p) . Note that a geometric random variable is just a negative binomial with parameter $(1, p)$.

In the next example, we use the negative binomial to obtain another solution of the problem of the points.

Example 8d

If independent trials, each resulting in a success with probability p , are performed, what is the probability of r successes occurring before m failures?

Solution The solution will be arrived at by noting that r successes will occur before m failures if and only if the r th success occurs no later than the $(r + m - 1)$ trial. This follows because if the r th success occurs before or at the $(r + m - 1)$ trial, then it must have occurred before the m th failure, and conversely. Hence, from Equation (8.2), the desired probability is

$$\sum_{n=r}^{r+m-1} \binom{n-1}{r-1} p^r (1-p)^{n-r} \quad \blacksquare$$

Example 8e**The Banach match problem**

At all times, a pipe-smoking mathematician carries 2 matchboxes—1 in his left-hand pocket and 1 in his right-hand pocket. Each time he needs a match, he is equally likely to take it from either pocket. Consider the moment when the mathematician first discovers that one of his matchboxes is empty. If it is assumed that both matchboxes initially contained N matches, what is the probability that there are exactly k matches, $k = 0, 1, \dots, N$, in the other box?

Solution Let E denote the event that the mathematician first discovers that the right-hand matchbox is empty and that there are k matches in the left-hand box at the time. Now, this event will occur if and only if the $(N + 1)$ choice of the right-hand matchbox is made at the $(N + 1 + N - k)$ trial. Hence, from Equation (8.2) (with $p = \frac{1}{2}$, $r = N + 1$, and $n = 2N - k + 1$), we see that

$$P(E) = \binom{2N-k}{N} \left(\frac{1}{2}\right)^{2N-k+1}$$

Since there is an equal probability that it is the left-hand box that is first discovered to be empty and there are k matches in the right-hand box at that time, the desired result is

$$2P(E) = \binom{2N-k}{N} \left(\frac{1}{2}\right)^{2N-k} \quad \blacksquare$$

Example 8f

Compute the expected value and the variance of a negative binomial random variable with parameters r and p .

Solution We have

$$\begin{aligned} E[X^k] &= \sum_{n=r}^{\infty} n^k \binom{n-1}{r-1} p^r (1-p)^{n-r} \\ &= \frac{r}{p} \sum_{n=r}^{\infty} n^{k-1} \binom{n}{r} p^{r+1} (1-p)^{n-r} \quad \text{since } n \binom{n-1}{r-1} = r \binom{n}{r} \\ &= \frac{r}{p} \sum_{m=r+1}^{\infty} (m-1)^{k-1} \binom{m-1}{r} p^{r+1} (1-p)^{m-(r+1)} \quad \text{by setting } m = n + 1 \\ &= \frac{r}{p} E[(Y-1)^{k-1}] \end{aligned}$$

where Y is a negative binomial random variable with parameters $r + 1, p$. Setting $k = 1$ in the preceding equation yields

$$E[X] = \frac{r}{p}$$

Setting $k = 2$ in the equation for $E[X^k]$ and using the formula for the expected value of a negative binomial random variable gives

$$\begin{aligned} E[X^2] &= \frac{r}{p} E[Y - 1] \\ &= \frac{r}{p} \left(\frac{r + 1}{p} - 1 \right) \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(X) &= \frac{r}{p} \left(\frac{r + 1}{p} - 1 \right) - \left(\frac{r}{p} \right)^2 \\ &= \frac{r(1 - p)}{p^2} \end{aligned} \quad \blacksquare$$

Thus, from Example 8f, if independent trials, each of which is a success with probability p , are performed, then the expected value and variance of the number of trials that it takes to amass r successes is r/p and $r(1 - p)/p^2$, respectively.

Since a geometric random variable is just a negative binomial with parameter $r = 1$, it follows from the preceding example that the variance of a geometric random variable with parameter p is equal to $(1 - p)/p^2$, which checks with the result of Example 8c.

Example 8g

Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

Solution Since the random variable of interest is a negative binomial with parameters $r = 4$ and $p = \frac{1}{6}$, it follows that

$$\begin{aligned} E[X] &= 24 \\ \text{Var}(X) &= \frac{4 \left(\frac{5}{6} \right)}{\left(\frac{1}{6} \right)^2} = 120 \end{aligned} \quad \blacksquare$$

8.3 The Hypergeometric Random Variable

Suppose that a sample of size n is to be chosen randomly (without replacement) from an urn containing N balls, of which m are white and $N - m$ are black. If we let X denote the number of white balls selected, then

$$P\{X = i\} = \frac{\binom{m}{i} \binom{N - m}{n - i}}{\binom{N}{n}} \quad i = 0, 1, \dots, n \quad (8.4)$$

A random variable X whose probability mass function is given by Equation (8.4) for some values of n, N, m is said to be a *hypergeometric* random variable.

Remark Although we have written the hypergeometric probability mass function with i going from 0 to n , $P\{X = i\}$ will actually be 0, unless i satisfies the inequalities $n - (N - m) \leq i \leq \min(n, m)$. However, Equation (8.4) is always valid because of our convention that $\binom{r}{k}$ is equal to 0 when either $k < 0$ or $r < k$. ■

**Example
8h**

An unknown number, say, N , of animals inhabit a certain region. To obtain some information about the size of the population, ecologists often perform the following experiment: They first catch a number, say, m , of these animals, mark them in some manner, and release them. After allowing the marked animals time to disperse throughout the region, a new catch of size, say, n , is made. Let X denote the number of marked animals in this second capture. If we assume that the population of animals in the region remained fixed between the time of the two catches and that each time an animal was caught it was equally likely to be any of the remaining uncaught animals, it follows that X is a hypergeometric random variable such that

$$P\{X = i\} = \frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}} \equiv P_i(N)$$

Suppose now that X is observed to equal i . Then, since $P_i(N)$ represents the probability of the observed event when there are actually N animals present in the region, it would appear that a reasonable estimate of N would be the value of N that maximizes $P_i(N)$. Such an estimate is called a *maximum likelihood* estimate. (See Theoretical Exercises 13 and 18 for other examples of this type of estimation procedure.)

The maximization of $P_i(N)$ can be done most simply by first noting that

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)}$$

Now, the preceding ratio is greater than 1 if and only if

$$(N-m)(N-n) \geq N(N-m-n+i)$$

or, equivalently, if and only if

$$N \leq \frac{mn}{i}$$

Thus, $P_i(N)$ is first increasing and then decreasing and reaches its maximum value at the largest integral value not exceeding mn/i . This value is the maximum likelihood estimate of N . For example, suppose that the initial catch consisted of $m = 50$ animals, which are marked and then released. If a subsequent catch consists of $n = 40$ animals of which $i = 4$ are marked, then we would estimate that there are some 500 animals in the region. (Note that the preceding estimate could also have been obtained by assuming that the proportion of marked animals in the region, m/N , is approximately equal to the proportion of marked animals in our second catch, i/n .) ■

**Example
8i**

A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

Solution Let A denote the event that the purchaser accepts a lot. Now,

$$\begin{aligned}
 P(A) &= P(A|\text{lot has 4 defectives})\frac{3}{10} + P(A|\text{lot has 1 defective})\frac{7}{10} \\
 &= \frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}}\left(\frac{3}{10}\right) + \frac{\binom{1}{0}\binom{9}{3}}{\binom{10}{3}}\left(\frac{7}{10}\right) \\
 &= \frac{54}{100}
 \end{aligned}$$

Hence, 46 percent of the lots are rejected. ■

If n balls are randomly chosen without replacement from a set of N balls of which the fraction $p = m/N$ is white, then the number of white balls selected is hypergeometric. Now, it would seem that when m and N are large in relation to n , it shouldn't make much difference whether the selection is being done with or without replacement, because, no matter which balls have previously been selected, when m and N are large, each additional selection will be white with a probability approximately equal to p . In other words, it seems intuitive that when m and N are large in relation to n , the probability mass function of X should approximately be that of a binomial random variable with parameters n and p . To verify this intuition, note that if X is hypergeometric, then, for $i \leq n$,

$$\begin{aligned}
 P\{X = i\} &= \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \\
 &= \frac{m!}{(m-i)!i!} \frac{(N-m)!}{(N-m-n+i)!(n-i)!} \frac{(N-n)!n!}{N!} \\
 &= \binom{n}{i} \frac{m}{N} \frac{m-1}{N-1} \cdots \frac{m-i+1}{N-i+1} \frac{N-m}{N-i} \frac{N-m-1}{N-i-1} \\
 &\quad \cdots \frac{N-m-(n-i-1)}{N-i-(n-i-1)} \\
 &\approx \binom{n}{i} p^i (1-p)^{n-i} \quad \begin{array}{l} \text{when } p = m/N \text{ and } m \text{ and } N \text{ are} \\ \text{large in relation to } n \text{ and } i \end{array}
 \end{aligned}$$

Example 8j

Determine the expected value and the variance of X , a hypergeometric random variable with parameters n , N , and m .

Solution

$$\begin{aligned}
 E[X^k] &= \sum_{i=0}^n i^k P\{X = i\} \\
 &= \sum_{i=1}^n i^k \binom{m}{i} \binom{N-m}{n-i} / \binom{N}{n}
 \end{aligned}$$

Using the identities

$$i \binom{m}{i} = m \binom{m-1}{i-1} \quad \text{and} \quad n \binom{N}{n} = N \binom{N-1}{n-1}$$

we obtain

$$\begin{aligned}
 E[X^k] &= \frac{nm}{N} \sum_{i=1}^n i^{k-1} \binom{m-1}{i-1} \binom{N-m}{n-i} / \binom{N-1}{n-1} \\
 &= \frac{nm}{N} \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{m-1}{j} \binom{N-m}{n-1-j} / \binom{N-1}{n-1} \\
 &= \frac{nm}{N} E[(Y+1)^{k-1}]
 \end{aligned}$$

where Y is a hypergeometric random variable with parameters $n-1$, $N-1$, and $m-1$. Hence, upon setting $k=1$, we have

$$E[X] = \frac{nm}{N}$$

In words, if n balls are randomly selected from a set of N balls, of which m are white, then the expected number of white balls selected is nm/N .

Upon setting $k=2$ in the equation for $E[X^k]$, we obtain

$$\begin{aligned}
 E[X^2] &= \frac{nm}{N} E[Y+1] \\
 &= \frac{nm}{N} \left[\frac{(n-1)(m-1)}{N-1} + 1 \right]
 \end{aligned}$$

where the final equality uses our preceding result to compute the expected value of the hypergeometric random variable Y .

Because $E[X] = nm/N$, we can conclude that

$$\text{Var}(X) = \frac{nm}{N} \left[\frac{(n-1)(m-1)}{N-1} + 1 - \frac{nm}{N} \right]$$

Letting $p = m/N$ and using the identity

$$\frac{m-1}{N-1} = \frac{Np-1}{N-1} = p - \frac{1-p}{N-1}$$

shows that

$$\begin{aligned}
 \text{Var}(X) &= np[(n-1)p - (n-1)\frac{1-p}{N-1} + 1 - np] \\
 &= np(1-p) \left(1 - \frac{n-1}{N-1} \right)
 \end{aligned}$$

Remark We have shown in Example 8j that if n balls are randomly selected without replacement from a set of N balls, of which the fraction p are white, then the expected number of white balls chosen is np . In addition, if N is large in relation to n [so that $(N-n)/(N-1)$ is approximately equal to 1], then

$$\text{Var}(X) \approx np(1-p)$$

In other words, $E[X]$ is the same as when the selection of the balls is done with replacement (so that the number of white balls is binomial with parameters n and p), and if the total collection of balls is large, then $\text{Var}(X)$ is approximately equal to what it would be if the selection were done with replacement. This is, of course, exactly what we would have guessed, given our earlier result that when the number of balls in the urn is large, the number of white balls chosen approximately has the mass function of a binomial random variable. ■

8.4 The Zeta (or Zipf) Distribution

A random variable is said to have a zeta (sometimes called the Zipf) distribution if its probability mass function is given by

$$P\{X = k\} = \frac{C}{k^{\alpha+1}} \quad k = 1, 2, \dots$$

for some value of $\alpha > 0$. Since the sum of the foregoing probabilities must equal 1, it follows that

$$C = \left[\sum_{k=1}^{\infty} \left(\frac{1}{k} \right)^{\alpha+1} \right]^{-1}$$

The zeta distribution owes its name to the fact that the function

$$\zeta(s) = 1 + \left(\frac{1}{2} \right)^s + \left(\frac{1}{3} \right)^s + \dots + \left(\frac{1}{k} \right)^s + \dots$$

is known in mathematical disciplines as the Riemann zeta function (after the German mathematician G. F. B. Riemann).

The zeta distribution was used by the Italian economist V. Pareto to describe the distribution of family incomes in a given country. However, it was G. K. Zipf who applied zeta distribution to a wide variety of problems in different areas and, in doing so, popularized its use.

9 Expected Value of Sums of Random Variables

A very important property of expectations is that the expected value of a sum of random variables is equal to the sum of their expectations. In this section, we will prove this result under the assumption that the set of possible values of the probability experiment—that is, the sample space S —is either finite or countably infinite. Although the result is true without this assumption (and a proof is outlined in the theoretical exercises), not only will the assumption simplify the argument, but it will also result in an enlightening proof that will add to our intuition about expectations. So, for the remainder of this section, suppose that the sample space S is either a finite or a countably infinite set.

For a random variable X , let $X(s)$ denote the value of X when $s \in S$ is the outcome of the experiment. Now, if X and Y are both random variables, then so is their sum. That is, $Z = X + Y$ is also a random variable. Moreover, $Z(s) = X(s) + Y(s)$.

Example 9a

Suppose that the experiment consists of flipping a coin 5 times, with the outcome being the resulting sequence of heads and tails. Suppose X is the number of heads in the first 3 flips and Y is the number of heads in the final 2 flips. Let $Z = X + Y$. Then, for instance, for the outcome $s = (h, t, h, t, h)$,

$$X(s) = 2$$

$$Y(s) = 1$$

$$Z(s) = X(s) + Y(s) = 3$$

meaning that the outcome (h, t, h, t, h) results in 2 heads in the first three flips, 1 head in the final two flips, and a total of 3 heads in the five flips. ■